

▼ Online Lecture Room + Office Hour Links**Lecture and Instructor Office Hours**https://canvas.wisc.edu/courses/478888/external_tools/27637**TA Office Hours** (https://canvas.wisc.edu/courses/478888/external_tools/27637)**▼ Lecture 1 - 09/04 - Introduction and Optical Illusions****Quiz - Introduction**

Sep 6 0.5 pts

**HW1**

Sep 16 5 pts

Slide**Lecture 1: Introduction and Optical Illusions** <https://drive.google.com/file/d/1YegzPtnTO1tYGmy0cHGGi7lCHoNYA6Qq/view?usp=sharing>**▼ Lecture 2 - 09/09 - Image Formation and Sensing****Quiz - Image Formation and Sensing**

Sep 11 0.5 pts

Videos

Video 1: Intro and Perspective Imaging



(<https://drive.google.com/file/d/1py9JP2ckzJsSO9aZmQH9AoauEPsmFFdX/view?usp=sharing>)

Video 2: Magnification and Vanishing Point



(<https://drive.google.com/file/d/1n2eW733G-YhODvgRDXiv-sJaiqxrRLJ/view?usp=sharing>)

Video 3: Lenses



(https://drive.google.com/file/d/1-R_qDzgspwc5_YCgjsi0V4NzeJt35wFO/view?usp=sharing)

Video 4: Defocus



(<https://drive.google.com/file/d/1gKStcg48QBy4OaEYU9oq6MyTEAdvJiEb/view?usp=sharing>)

Video 5: Lens-based Image Artifacts



(<https://drive.google.com/file/d/1aJnEWKtnKHyTBjeR-UAbg76kTegQjHWD/view?usp=sharing>)

Video 6: Image Sensors, Noise, and Dynamic Range



(<https://drive.google.com/file/d/1z4V4iFkacDUbOYx8i0gouQLcW1oN3XgM/view?usp=sharing>)

Slide

Lecture 2: Image Formation and Sensing



(<https://drive.google.com/file/d/19ROy1qZgFq9nQPu3PMMmTGIXiT7iOEPm/view?usp=sharing>)

▼ Lecture 3 - 09/11 - Image Filtering

Quiz - Image Filtering



Sep 13 0.5 pts

Videos

Video 1: Intro



(https://drive.google.com/file/d/1IQVt0OEUUqeVjpfGRNeyjG_6j64UW52N/view?usp=sharing)

Video 2: Linear Filtering



(<https://drive.google.com/file/d/1kD7dDewYppHl3y4Q9sGjGSrtfApzgJl/view?usp=sharing>)

Video 3: Box Filter Example Linear Filtering



(https://drive.google.com/file/d/1dH3nDw_a6DW6S66PoGRSIK22Of7uo2t8/view?usp=sharing)

Video 4: Filtering Examples and Gaussian Smoothing



(https://drive.google.com/file/d/1Fu8mF9MQ2amubzK60FO_G_hU0NEiJE71/view?usp=sharing)

Video 5: Median Filtering



(<https://drive.google.com/file/d/1HEPGwRik2HYCw11ybZmXdRMBLGgTXv8A/view?usp=sharing>)

Video 6: Template Matching



(<https://drive.google.com/file/d/1nWczVoshrrNipIgrRXsIL5kL8DAtdD/view?usp=sharing>)

Slide

Lecture 3: Image Filtering



(https://drive.google.com/file/d/10Sstb1q2z80PvUyH_iZhYXil_vPLh8he/view?usp=sharing)

▼ Lecture 4 - 09/16 - Thinking about Images in the Fourier Domain



Quiz -Thinking about Images in the Fourier Domain

Sep 18 0.5 pts

Videos



Video 1: Intro to Fourier Transform

(<https://drive.google.com/file/d/1YrWmW9SRYs4O7d8emnKfhMICFPIU-nTf/view?usp=sharing>)



Video 2: Fourier Transform Examples

(<https://drive.google.com/file/d/1F9UqAGmrmxZ-3VU4xqL9plqDm4hSu7DH/view?usp=sharing>)



Video 3: Fourier Transform Properties

(<https://drive.google.com/file/d/1OlnZqdfp7SuvrvduPqlrH4EP2BDFhAXF/view?usp=sharing>)



Video 4: 2D Fourier Transform Example (<https://drive.google.com/file/d/1hiN-kacix-okd3aCWOxpgWHQj55Seqe6/view?usp=sharing>)



Video 5: Filtering in the Fourier Domain

(<https://drive.google.com/file/d/1PVXvU38eA2yVdDj8g3pN74h72B0Err8d/view?usp=sharing>)



Video 6: Deconvolution (https://drive.google.com/file/d/1WI_AXvAdTxM-IS5ko2AB9tZPD9tcmK-V/view?usp=sharing)



Video 7: Sampling (https://drive.google.com/file/d/1OF89U-aXdAku6e31clag_4nYWbiR_8yg/view?usp=sharing)

Slide



Lecture 4: Thinking about Images in the Fourier Domain

(<https://drive.google.com/file/d/1XA96oILFhR5XZ5RuniYB8li68KEfFV25/view?usp=sharing>)

▼ Lecture 5 - 09/18 - Edge and Corner Detection



Quiz - Edge and Corner Detection

Sep 20 0.5 pts

Videos



Video 1: Intro (<https://drive.google.com/file/d/1JdWQJ5oYv-7Thf8mOEUK2qK8V6CuS4qm/view?usp=sharing>)



Video 2: Theory of Gradient Edge Detector (<https://drive.google.com/file/d/1zU48w4P2K2AZAm1uSS7Rg-6J4vZz3coP/view?usp=sharing>)



Video 3: Discrete Gradient Operator (https://drive.google.com/file/d/1qp_y3xr833CW5Gbn9XDDnkPFK0EJxd18/view?usp=sharing)



Video 4: Laplacian and Noise Suppression (<https://drive.google.com/file/d/1leePnVUHpylqLSFR1g1FtENOzwqpqMZ7/view?usp=sharing>)



Video 5: Canny Edge Detector (<https://drive.google.com/file/d/1C39Wh8voTLCFEh9rTZJSpsQCOZB7jK44/view?usp=sharing>)



Video 6: Harris Corner Detection (https://drive.google.com/file/d/10PksSdFL-tdhU7iGZsscmK_D8j-3Jiy0/view?usp=sharing)

Slide



Lecture 5: Edge and Corner Detection (<https://drive.google.com/file/d/1JBBDRxwbwQeO1tNzGXgnVOX7wjWPMLC-/view?usp=sharing>)

▼ Lecture 6 & Lecture 7 - 09/23 & 9/25 - Boundary Detection and Hough Transforms & Video ...**HW2**

Oct 7 16 pts

**Quiz - Image Formation, Processing, and Edge Detection Review**

Sep 25 0.5 pts

**Quiz - Boundary Detection and Hough Transforms**

Sep 27 0.5 pts

Lecture Videos**Video 1: Intro** https://drive.google.com/file/d/1rDbK2V5_zdRYsjUYgaUT33G1EATf7qQv/view?usp=sharing**Video 2: Line Fitting** <https://drive.google.com/file/d/1-BdIUcnT-9sP90LUIY3MYVTEajVQ2RYyG/view?usp=sharing>**Video 3: Hough Line** <https://drive.google.com/file/d/1z17-BI71H2q3gPcrXPkspVsTUoMbEkAB/view?usp=sharing>**Video 4: Hough Transform Practical Considerations** <https://drive.google.com/file/d/1EHIJHb4iEHZm1cMG--m-5KibXZqwqWvs/view?usp=sharing>**Video 5: Circle and Generalized Hough Transform** <https://drive.google.com/file/d/1fqSLV4tnGaH6N749iNr1YlkdY2KT8Gtd/view?usp=sharing>**Video 6: Active Contours Intro** https://drive.google.com/file/d/1zDjyzf22VocZCFHI_1a20LFi9wHe_Q7-/view?usp=sharing**Video 7: Active Contours Optimization** <https://drive.google.com/file/d/1rrnLa4xbP7DoXub0T8kjN2Fr4p4ernv5/view?usp=sharing>

Slide

[Lecture 6 & 7 Boundary Detection and Hough Transforms & Video Boundary](#)

[Detection using Active Contours](#) 

 (<https://drive.google.com/file/d/1dj8SWS3zoNS2EhJ1IjVcinHLZVxo066O/view?usp=sharing>)

▼ Lecture 8 - 09/30 - 2D Object Recognition using SIFT

No Quiz for Lecture 8

Videos

[Video 1: Intro to Image Matching](#) 

 (https://drive.google.com/file/d/1EYEUlAcR8svX_FbcX6oAYYTZhCcmicB-/view?usp=sharing)

[Video2: 1D Blob Detection](#) 

 (<https://drive.google.com/file/d/1F8rm4xOnxf7KbQl9PKpZjfUt3Uut8OiZ/view?usp=sharing>)

[Video 3: 2D Blob Detection](#) 

 (<https://drive.google.com/file/d/1TvDNYffld0GwZp49WKVet7Tyq4ZjTdCJ/view?usp=sharing>)

[Video 4: Scale Space Analysis](#) 

 (<https://drive.google.com/file/d/11hsay12JVJ0R6X50r9oYxJE8o8WiLpDi/view?usp=sharing>)

[Video 5: SIFT Descriptor](#)  (https://drive.google.com/file/d/17GZyKVjGW2LCeYd-XJLKNLgNmEK_-Bm_/view?usp=sharing)



[Video 6: SIFT Examples and Active Contours](#) 

 (https://drive.google.com/file/d/1uiv2_iLSnTYM-gcOMJSKqly3-D1q-ARs/view?usp=sharing)

Slide

 [Lecture 8 2D Object Recognition using SIFT](https://drive.google.com/file/d/1oUQLi-cMmjQIS7vtfhLADhyorF-tR1hn/view?usp=sharing)  [_](https://drive.google.com/file/d/1oUQLi-cMmjQIS7vtfhLADhyorF-tR1hn/view?usp=sharing) (https://drive.google.com/file/d/1oUQLi-cMmjQIS7vtfhLADhyorF-tR1hn/view?usp=sharing)

▼ Lecture 09 and 10 - 10/02 and 10/07 - Image Alignment: Image Transformations and RANSAC

 **HW3**
Oct 23 13 pts

 **Survey about Quizzes**
Dec 8

 **Quiz - Image Alignment**
Oct 4 0.5 pts

Videos

 [Video 1: Intro and Overview](https://drive.google.com/file/d/1jplfVRR6ZFGPZpZcXAGPhIdHQECDYzVm/view?usp=sharing)  [_](https://drive.google.com/file/d/1jplfVRR6ZFGPZpZcXAGPhIdHQECDYzVm/view?usp=sharing) (https://drive.google.com/file/d/1jplfVRR6ZFGPZpZcXAGPhIdHQECDYzVm/view?usp=sharing)

 [Video 2: Linear Warping](https://drive.google.com/file/d/1SMOteesTEw-NY_tGoeMeQazHp0AuJvXF/view?usp=sharing)  [_](https://drive.google.com/file/d/1SMOteesTEw-NY_tGoeMeQazHp0AuJvXF/view?usp=sharing) (https://drive.google.com/file/d/1SMOteesTEw-NY_tGoeMeQazHp0AuJvXF/view?usp=sharing)

 [Video 3: Homogeneous Coordinates](https://drive.google.com/file/d/1I0PSecYHr9fN1gLh5-wdxxMOvYeoZKeS/view?usp=sharing)  [_](https://drive.google.com/file/d/1I0PSecYHr9fN1gLh5-wdxxMOvYeoZKeS/view?usp=sharing) (https://drive.google.com/file/d/1I0PSecYHr9fN1gLh5-wdxxMOvYeoZKeS/view?usp=sharing)

 [Video 4: Homography](https://drive.google.com/file/d/18VWBnM0F13zkUz3qQRT5nHEmr0CBvr8V/view?usp=sharing)  [_](https://drive.google.com/file/d/18VWBnM0F13zkUz3qQRT5nHEmr0CBvr8V/view?usp=sharing) (https://drive.google.com/file/d/18VWBnM0F13zkUz3qQRT5nHEmr0CBvr8V/view?usp=sharing)

Video 5: Least Squares Homography  (https://drive.google.com/file/d/1vFXPOR646b9TxEmrxV_Wo9fSWtq0PPgC/view?usp=sharing)**Video 6: RANSAC**  (https://drive.google.com/file/d/1z48_p8INvSYZ3QJ45NdOuGsBowY1e_-u/view?usp=sharing)**Video 7: Warping**  (https://drive.google.com/file/d/1ayn_v57W7FxxPF1TKvZ_g47LfyqXFt9y/view?usp=sharing) **Video 8: Blending**  (<https://drive.google.com/file/d/1aFC-mmhtphMpRb7uMf5NGY1HRb-y6BV-/view?usp=sharing>)**Slide****Lecture 9 & 10 Image Alignment: Image Transformations and RANSAC**  (<https://drive.google.com/file/d/1zCpTBA26FtLFKfuvHNVmoG3oLerULMFT/view?usp=sharing>)**▼ Lecture 11 - 10/09 - Face Detection with Support Vector Machines** **Quiz - Face Detection and SVM**

Oct 14 0.5 pts

Videos **Video 1: Intro to Face Detection** **Video 2: Haar Features**



Video 3: Integral Images



Video 4: Nearest Neighbor Classifier



Video 5: Linear Decision Boundaries



Video 6: Support Vector Machines

Slide



Lecture 11 Face Detection with Support Vector Machines

(https://drive.google.com/file/d/1yLJH0080tZCTt3TC2UNk8t_N3aFm8PNr/view?usp=sharing)

▼ Lecture 12 - 10/14 - Image Segmentation with k-Means Clustering and Mean-Shift

No Quiz for this Lecture

Videos



Video 1: Intro

(<https://drive.google.com/file/d/1uk0Nb3jiNM0LeWzUMb10XHx5x7egyGig/view?usp=sharing>)



Video 2: Clustering

(<https://drive.google.com/file/d/1FLS8b9t5oWU02m3memHO46wRBzGLNrTb/view?usp=sharing>)



Video 3: K-Means Clustering

(https://drive.google.com/file/d/19ShI7vkYIIOMJe_IFbz1_Vx-XqzG_9Eo/view?usp=sharing)



Video 4: K Means Segmentation (<https://drive.google.com/file/d/1nPP1C-UbDBtjM42J84qnsFZhxiB0NsEM/view?usp=sharing>).



Video 5: Mean Shift Clustering (<https://drive.google.com/file/d/1iq1oINrryzumEN0y0IM8hENvCkvzPrqg/view?usp=sharing>).



Video 6: Mean Shift Segmentation (<https://drive.google.com/file/d/14oVnc-cOC1wAM2NNHPJTkrImoKduwdzl/view?usp=sharing>).



Video 7: Graph Cut Segmentation (<https://drive.google.com/file/d/1KdM5HIWzK4xD2TtkaQ0RISD45WxtdUog/view?usp=sharing>).



Video 8: Normalized Cut Segmentation (<https://drive.google.com/file/d/1Z70gxD7u1BmHXdOs-SeejAD3ajkeInV9/view?usp=sharing>).

Slide



Lecture 12 Image Segmentation with k-Means Clustering and Mean-Shift (<https://drive.google.com/file/d/1yUMqEtEL9ZfB-e83pvFylrpBBQgzcUd-/view?usp=sharing>).

▼ Lecture 13 - 10/16 - Introduction to Neural Networks and Convolutional Neural Networks



Quiz - Introduction to Neural Networks

Oct 22 0.5 pts

Videos

01 - Introduction



<https://drive.google.com/file/d/1zDztKoNIMhFXrKSNTe5cRhLmJc4eVVHE/view?usp=sharing>

02 - Neurons and Activations



<https://drive.google.com/file/d/1ton280boWNkVfCRb7EWDjWV7NtuusW9a/view?usp=sharing>

03 - Character Recognition Example



<https://drive.google.com/file/d/1dh2nd6Gznl50qDrXkhlyGhaEon5DJ0Ck/view?usp=sharing>

04 - Neural Network Training



<https://drive.google.com/file/d/1gwaoTtF97Wsd8T43r2JmyL8xPgyVtjPF/view?usp=sharing>

05 - Motivating Convolutional Networks https://drive.google.com/file/d/1-lvWVWYCR4PFmMlvkkg_X3bdu4aA7Fc4/view?usp=sharing



06 - Convolutional Networks Description



<https://drive.google.com/file/d/1TXcKYhPTMHJwQS0a0KUIXiyFOsCsGCUK/view?usp=sharing>

07 - ConvNets for Semantic Segmentation



https://drive.google.com/file/d/1fKZX_LVX5kvE4CpCm6PlkU1NqsF3DIkZ/view?usp=sharing

Slide

Lecture 13 Intro to Neural Networks and Convolutional Neural Networks



https://drive.google.com/file/d/1H3wXPHw0_Dwo6K36cIlm9LsJ4Sud6U9a/view?usp=sharing

▼ Lecture 14 & 15- 10/23 & 10/28 - Radiometry and Surface Appearance & 3D Shape from Pho...



Quiz - Radiometry and Photometric Stereo

Oct 30 0.5 pts

Videos



01 - Introduction to Radiometry (https://drive.google.com/file/d/1ZlO38-K9eL_zq-tXitYwz_juJ7bTQF4D/view?usp=sharing)



02 - BRDF (https://drive.google.com/file/d/1BwVWV5Ymx_akzYsx0-AjH6FYqrDEA5nA/view?usp=sharing)



03 - Lambertian Model (<https://drive.google.com/file/d/1LqGLbqRxtD8mSDkGJURk1D9mfGqVDw0M/view?usp=sharing>)



04 - Specular and Rough Surfaces (<https://drive.google.com/file/d/1CGA2htWEaaNGqwPHG2xyc4KshbVneggp/view?usp=sharing>)



05 - Photometric Stereo (https://drive.google.com/file/d/1z-Qz9bJaswW_q4ixNi6xs7qNjwZnhQbl/view?usp=sharing)



06 - Photometric Stereo Analysis and Applications (https://drive.google.com/file/d/19_76YApJOc5HHwPPrZ3nmS6KzC__rCZ1/view?usp=sharing)



07 - Example-based Photometric Stereo (<https://drive.google.com/file/d/1HZFJBAFRdl7lxdCzMypEzXtlflo652Qf/view?usp=sharing>)

Slide

 [Lecture 14 15 Radiometry and Surface Appearance 3D Shape from Photometric Stereo](https://drive.google.com/file/d/1jRCzU5bHWHfQmqij8XiTK3WJfprL0Dsg/view?usp=sharing)  (<https://drive.google.com/file/d/1jRCzU5bHWHfQmqij8XiTK3WJfprL0Dsg/view?usp=sharing>)

▼ Lecture 16 - 10/30 - Shape from Focus/Defocus

 **Quiz - Shape from Focus/Defocus**
Nov 4 0.5 pts

 **HW4**
Nov 13 7 pts

Videos

 [01 - Introduction to Camera Defocus](https://drive.google.com/file/d/1qfa9HD2jtTkV6vYKXiBBqgee4jUkQ8sl/view?usp=sharing)  (<https://drive.google.com/file/d/1qfa9HD2jtTkV6vYKXiBBqgee4jUkQ8sl/view?usp=sharing>)

 [02 - Depth from Focus](https://drive.google.com/file/d/1rB1RsLKcE82Dh2zGDy-Cm7iE-MxXf7gu/view?usp=sharing)  (<https://drive.google.com/file/d/1rB1RsLKcE82Dh2zGDy-Cm7iE-MxXf7gu/view?usp=sharing>)

 [03 - Depth from Defocus](https://drive.google.com/file/d/1PrC-1EMr18Wwl99hKnUbgSKPGKGnK4le/view?usp=sharing)  (<https://drive.google.com/file/d/1PrC-1EMr18Wwl99hKnUbgSKPGKGnK4le/view?usp=sharing>)

Slide

 [Lecture 16 Shape from Focus/Defocus](https://drive.google.com/file/d/1Zb-rNSK5WWSTGTaA7D_GWNY04VgO7vOx/view?usp=sharing)  (https://drive.google.com/file/d/1Zb-rNSK5WWSTGTaA7D_GWNY04VgO7vOx/view?usp=sharing)

▼ Lecture 17 - 11/04 - Camera Calibration and Shape from Stereo



Quiz - Camera Calibration and Simple Stereo

Nov 11 0.5 pts

Videos



[01 - Introduction to Depth from Stereo](#)

(https://drive.google.com/file/d/1iXVVfBu3hxVWcVYcINd2iRT2h_hRS-2j/view?usp=sharing)



[02 - Intrinsic Camera Matrix](#)

(<https://drive.google.com/file/d/1FDcg6OH1wHQx3rxzYmAVj9ICa0COoCcd/view?usp=sharing>)



[03 - Extrinsic Camera Matrix](#)

(https://drive.google.com/file/d/1IJKqi_Gx1sAXAKV_RHCQy8VSE1KoMNPq/view?usp=sharing)



[04 - Camera Calibration](#)

(<https://drive.google.com/file/d/1AI7eEILBPf-RWZ7dRoA1cWKYuqK8-Ff3/view?usp=sharing>)



[05 - Recovering Extrinsic and Intrinsic Matrices](#)

(https://drive.google.com/file/d/1ItWqA8w4VMS5qVbjXTZ6_x2PrM0eWKwF/view?usp=sharing)



[06 - Triangulation](#)

(https://drive.google.com/file/d/1axNE8xQdAtXhJX-SUimgArHJI7nZ_Ooo/view?usp=sharing)



[07 - Finding Correspondences](#)

(https://drive.google.com/file/d/13Wpk1dbnIA3fbb9W_84Wa5OkI68vMesX/view?usp=sharing)

Slide

[Lecture 17 Camera Calibration and Shape from Stereo](#)



https://drive.google.com/file/d/1Z_JQWDPbO7vomjBc8qV6xA2bVXO5YuHQ/view?usp=sharing

▼ Lecture 18 - 11/06 - Internet-Scale 3D Reconstruction



Quiz - Internet-Scale 3D Reconstruction

Nov 11 0.5 pts

Videos

[01 - Introduction and Review of Basic Stereo](#)



https://drive.google.com/file/d/1FlfMa9R7yXqh9L_5MYbxx8F6DWqJIUiX/view?usp=sharing

[02 - Intuition and Outline](#)



https://drive.google.com/file/d/1-HABfFY2WDdSE8bA54n5dO_Y1I9K6l2x/view?usp=sharing

[03 - Essential Matrix](#)



<https://drive.google.com/file/d/15gEN8WP4nBiEOQTiaM5LHLTGiz8qt241/view?usp=sharing>

[04 - Fundamental Matrix](#)



<https://drive.google.com/file/d/1U68XSNGguKjv-rzvGjfeXWewDy6RWxMy/view?usp=sharing>

[05 - Stereo Autocalibration](#)



https://drive.google.com/file/d/1R7eQvzZ-5hF9uWkrKljGu_9-_GEx7aFt/view?usp=sharing

06 - Finding Dense Correspondences via Epipolar Geometry



(<https://drive.google.com/file/d/1vq3tVERU9EZsOamq4-EGjC8dM9D8lbJq/view?usp=sharing>)

07 - Triangulation and Internet Scale 3D



(<https://drive.google.com/file/d/12eUmpL3jPd2uyhOec6P9-AotJ2VmGY4d/view?usp=sharing>)

Slide

Lecture 18 Internet-Scale 3D Reconstruction



(https://drive.google.com/file/d/1Z_0GW7me7uAT6P0vVjfGnkejWm24F7nA/view?usp=sharing)

▼ Lecture 19 - 11/11 - Optical Flow and Motion Measurement



Quiz - Optical Flow and Motion

Nov 13 0.5 pts



HW5

Dec 2 14 pts

Videos

01 - Introduction to Motion Estimation



(<https://drive.google.com/file/d/1HWhs6xQQczwGZs--hiC7gWQOBRIPa4WB/view?usp=sharing>)

02 - Theory of Optical Flow (<https://drive.google.com/file/d/1R-i7ZWEBNW7k4Lm7MBTM5IAINxwm0pFU/view?usp=sharing>)



03 - Lucas-Kanade Optical Flow (<https://drive.google.com/file/d/1dCCGcp7zC4IE1-bE4aGYnYjLiS8hmdmA/view?usp=sharing>)





04 - Large-Motion Optical Flow (<https://drive.google.com/file/d/1-DTEdQVWmmmoWNRVnu20QLB2vHm7-HPg/view?usp=sharing>).

Slide



Lecture 19 Optical Flow and Motion Measurement
(https://drive.google.com/file/d/1UN1WH2L09HZequlcy_AX5bUMpmEq7hC_/view?usp=sharing).

▼ Lecture 20 - 11/13 - Background Subtraction and Image Tracking



HW5

Dec 2 14 pts



Quiz - Object Tracking

Nov 18 0.5 pts

Videos



01 - Introduction and Change Detection
(<https://drive.google.com/file/d/1QwkXAL5ZRTqf9Mnkzn4n9N0ngwUzYfSR/view?usp=sharing>).



02 - Motion Detection using Gaussian Mixture Models
(<https://drive.google.com/file/d/1x4fSxxJIQAJF4yfkMtGxYHIVuEkeQs4S/view?usp=sharing>).



03 - Template-based Tracking (<https://drive.google.com/file/d/1I-sNNgglwPiSXniOEEOsZLVHpYjTlaUN/view?usp=sharing>).

04 - Detection-based Tracking



(<https://drive.google.com/file/d/1QyIHoyLYu78HTKht8v3TAgOVIj2TKSFh/view?usp=sharing>)

Slide

Lecture 20 Background Subtraction and Image Tracking



(<https://drive.google.com/file/d/1ZUPqOlu3QqMSIKCmaKXtD09zhtlo5Gji/view?usp=sharing>)

▼ Lecture 21 - 11/18 - Modern 3D Cameras: Structured Light and Time-of-Flight



Quiz - Structured Light and Time-of-Flight

Nov 20 0.5 pts

Videos

01 - Introduction of Structured Light and Time-of-Flight



(<https://drive.google.com/file/d/1i7ugpJinFg7RD4G2SsMsKTxdo3tThjuW/view?usp=sharing>)

02 - Active Stereo



(<https://drive.google.com/file/d/1n6-SGwLSf8N1XDauJT7L3XLWvRyR6UF9/view?usp=sharing>)

03 - Structured Light Basics



(<https://drive.google.com/file/d/1PvJFEom7Zaoi5flmN0ozInjA476sHQBg/view?usp=sharing>)

04 - Coded Structured Light



(https://drive.google.com/file/d/1O_2kZLbIkDYkRWxJdT2Sx2IsGAMUcm9U/view?usp=sharing)

05 - Time of Flight Basics



(<https://drive.google.com/file/d/162uqk0sZArhVrNK2ozpNpvZMCmqcLUos/view?usp=sharing>)

06 - Continuous Wave Time of Flight

 (<https://drive.google.com/file/d/14oD3pT7vfM6kGYNnFfSEL6fcdCjMpmlB/view?usp=sharing>)

Slide

Lecture 21 Modern 3D Cameras Structured Light and Time of Flight

 (<https://drive.google.com/file/d/1dFVhmke7E2ovCwE7NjQu-1sFUQVfGbpc/view?usp=sharing>)

▼ Lecture 22 - 11/20 - Advanced Topics: Computational Cameras